

CSE400 – Lecture 7 Lecture Scribe

Expectation, CDFs, PDFs and Problem Solving

1 Lecture Coverage (From Outline Slide)

The lecture includes:

1. Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Example
2. Probability Density Function (PDF)
 - Definition
 - Properties
 - Example
3. Expectation of Random Variables
 - Definition and example
 - Expectation of a function of RV
 - Linear operation with expectation
 - n-th moments and central moments (Variance, Skewness, Kurtosis)

2 Cumulative Distribution Function (CDF)

2.1 Definition and Notation

Let X be a random variable.

The cumulative distribution function (CDF) is defined as:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Notation

X : random variable

$F_X(x)$: cumulative distribution function of X

Statement from slides:

Most information about the random experiment described by X is determined by the behavior of $F_X(x)$.

2.2 Assumptions and Conditions (Properties of a CDF)

A function must satisfy the following conditions to be a valid CDF.

1. Bounds

$$0 \leq F_X(x) \leq 1$$

2. Limits

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

3. Monotonicity

If $x_1 < x_2$, then

$$F_X(x_1) \leq F_X(x_2)$$

4. Probability from CDF

For $x_1 < x_2$,

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

3 Worked Example 1 — Validity of CDF

Problem: Determine which functions are valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step verification

Step 1: Check limits.

As $x \rightarrow -\infty$,

$$\tan^{-1}(x) \rightarrow -\frac{\pi}{2}$$

$$F_X(-\infty) = \frac{1}{2} - \frac{1}{2} = 0$$

As $x \rightarrow \infty$,

$$\tan^{-1}(x) \rightarrow \frac{\pi}{2}$$

$$F_X(\infty) = \frac{1}{2} + \frac{1}{2} = 1$$

Step 2: Since $\tan^{-1}(x)$ is increasing, the function is non-decreasing.

Step 3: Values remain between 0 and 1.

Valid CDF.

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Step-by-step verification

Step 1: For $x < 0$,

$$u(x) = 0 \Rightarrow F_X(x) = 0$$

Step 2: For $x \geq 0$,

$$F_X(x) = 1 - e^{-x}$$

Step 3: Limits:

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Step 4: Function increases with x .

Valid CDF.

(c)

$$F_X(x) = e^{-x^2}$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow 0$$

which violates $F_X(\infty) = 1$.

Invalid CDF.

(d)

$$F_X(x) = x^2 u(x)$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow \infty$$

which violates $0 \leq F_X(x) \leq 1$.

Invalid CDF.

4 Worked Example 2 — Computing Probabilities from a CDF

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a)

$$\Pr(X > 5)$$

Step 1:

$$\Pr(X > 5) = 1 - \Pr(X \leq 5)$$

Step 2:

$$= 1 - F_X(5)$$

Step 3:

$$= 1 - (1 - e^{-5})$$

Result:

$$= e^{-5}$$

(b)

$$\Pr(X < 5)$$

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c)

$$\Pr(3 < X < 7)$$

Step 1:

$$\Pr(3 < X < 7) = F_X(7) - F_X(3)$$

Step 2:

$$= (1 - e^{-7}) - (1 - e^{-3})$$

Result:

$$= e^{-3} - e^{-7}$$

(d) Conditional Probability

$$\Pr(X > 5 \mid X < 7)$$

Step 1:

Let

$$A = \{X > 5\}, \quad B = \{X < 7\}$$

Step 2:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr(A \cap B) = \Pr(5 < X < 7) = F_X(7) - F_X(5)$$

$$\Pr(B) = F_X(7)$$

Step 3:

Final expression:

$$\Pr(X > 5 \mid X < 7) = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

5 Probability Density Function (PDF)

5.1 Definition

For a continuous random variable X ,

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

5.2 Relationship Between PDF and CDF (Step-by-Step Derivation)

From the slides:

For a continuous range,

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into PDF definition:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

By definition of derivative,

$$f_X(x) = \frac{d}{dx} F_X(x)$$

5.3 Result (Converse Statement)

The CDF can be written as the integral of the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

6 Expectation of Random Variables (As Listed in Slides)

The lecture includes:

Definition and example of expectation

Expectation of a function of a random variable

Linear operation with expectation
n-th moments and central moments:

- Variance
- Skewness
- Kurtosis

(Only topics listed in slides are included; no additional formulas introduced.)

End of Lecture Scribe

This lecture scribe is:

Fully restricted to lecture slide content

Structured for exam revision

Step-by-step where derivations appear in slides

Free from external intuition or added explanations

If required, another independent version with a different structural ordering can be generated while remaining fully slide-faithful.